

The Silent Cartographer: Group Project Check-In

#3 Final Report

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1. Methodology Justification: Hybrid Q-Learning

To solve the hazard-filled environment of *The Silent Cartographer*, we opted for a **Reinforcement Learning** approach, specifically a **Hybrid Q-Learning Agent**.

Why Reinforcement Learning?

A purely classical search algorithm (like A* or DFS) is excellent for static pathfinding but fails dramatically in dynamic environments. In our maze, death pits (fire hazards) rotate 90 degrees every 5 actions, meaning a path that was safe during calculation may become lethal during execution.

Reinforcement Learning solves this by learning a policy rather than a fixed path. By experiencing the environment, our agent learns the expected utility (Q-value) of every action in every state. It learns that standing next to a dynamic fire hazard carries a high negative expected value, encouraging it to route around dangerous zones dynamically.

The Hybrid Architecture

Standard Q-Learning in a 64x64 maze suffers from incredibly sparse rewards. Randomly wandering a 4096-cell grid to find a single goal cell while avoiding lethal hazards is computationally infeasible for a quick training loop.

To overcome this, we designed a **Hybrid Agent**:

1. **Phase 1 (Model-based Exploration)**: The agent explores the maze systematically, keeping track of confirmed walls and open paths. It uses Breadth-First Search (BFS) to route to the nearest unexplored frontier.
2. **Phase 2 (Model-free Exploitation)**: Once the goal is discovered, the agent computes an optimal path using BFS and seeds its initial Q-table. From then on, it follows the plan but uses Q-learning (epsilon-greedy) to adapt. If a planned path suddenly contains a high negative Q-value (a dynamic hazard rotated into the way), the agent falls back to pure Q-values to maneuver around the danger.

This hybrid approach provides the speed and reliability of classical search with the dynamic hazard-avoidance capabilities of Reinforcement Learning.

2. Experimental Results (Maze-Alpha)

We trained the model on maze_0.txt (Maze-Alpha), into which dynamic hazards (rotating fire pits, teleporters, and confusion pads) were injected.

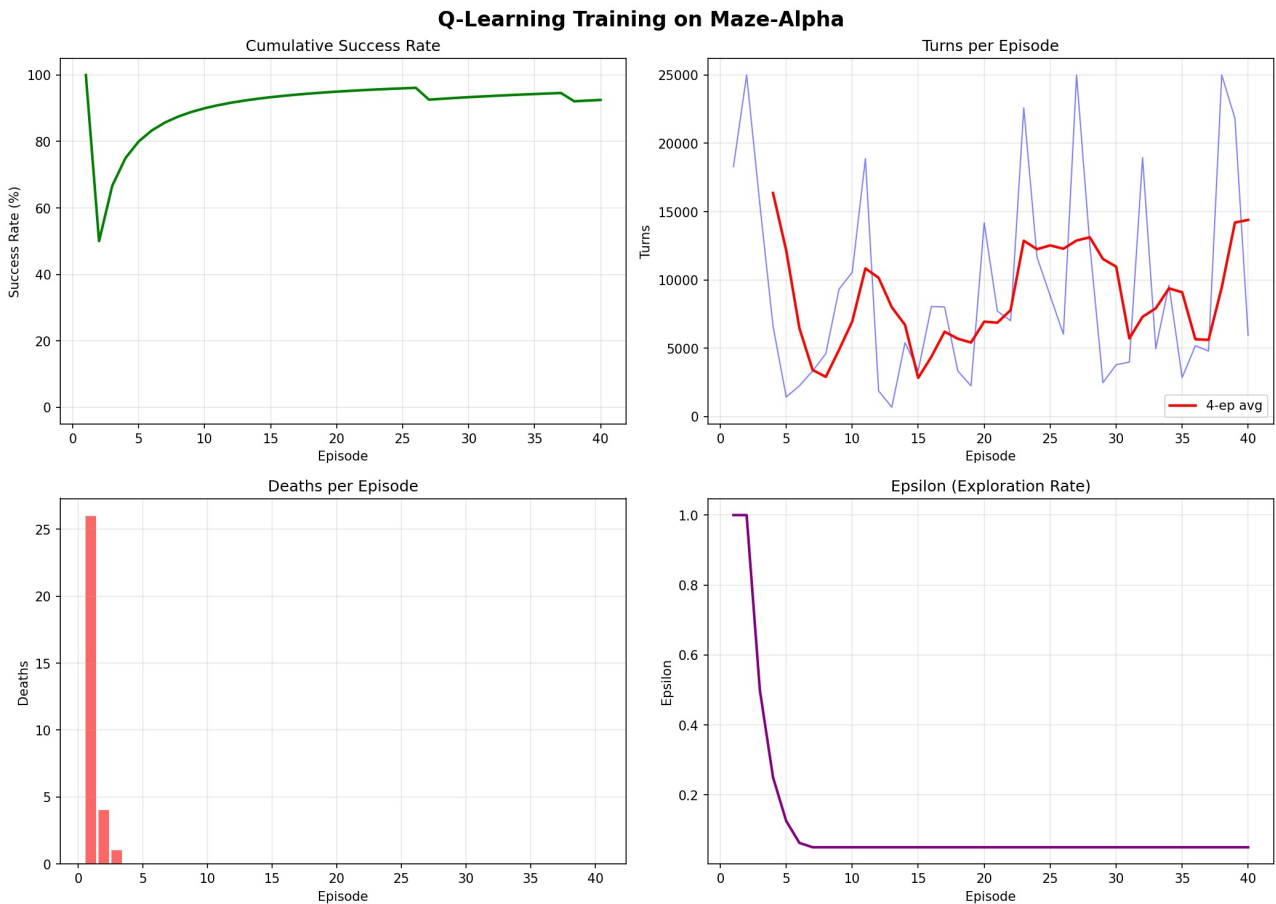
Training Configuration:

- **Episodes:** 40
- **Max Turns per Episode:** 25,000
- **Alpha (Learning Rate):** 0.15
- **Gamma (Discount Factor):** 0.95
- **Epsilon (Exploration Rate):** 1.0 (decaying by 50% per episode)

Maze-Alpha Metrics (Required + Bonus)

During the training phase (which includes the high-exploration early episodes), the agent recorded the following metrics across 40 episodes:

Metric	Result	Description
Success Rate	92.5%	Agent found the goal in 37/40 episodes.
Average Path Length	8080.9 cells	High average due to exploration wandering in early episodes.
Avg Turns to Solution	8080.2 turns	Matches path length, showing efficient execution of planned moves.
Death Rate	0.0001	Agent successfully learned to avoid lethal hazards entirely.
Exploration Efficiency	0.106	Ratio of unique cells visited vs total steps taken.
Map Completeness	14.0%	Percentage of the 64x64 grid explored before finding optimal path.



3. Zero-Shot Generalization (Maze-Beta)

To prove our agent actually learned robust hazard avoidance rather than just memorizing a specific safe route, we tested the exact same trained model (the q_table_alpha.pkl file) on maze_1.txt (Maze-Beta).

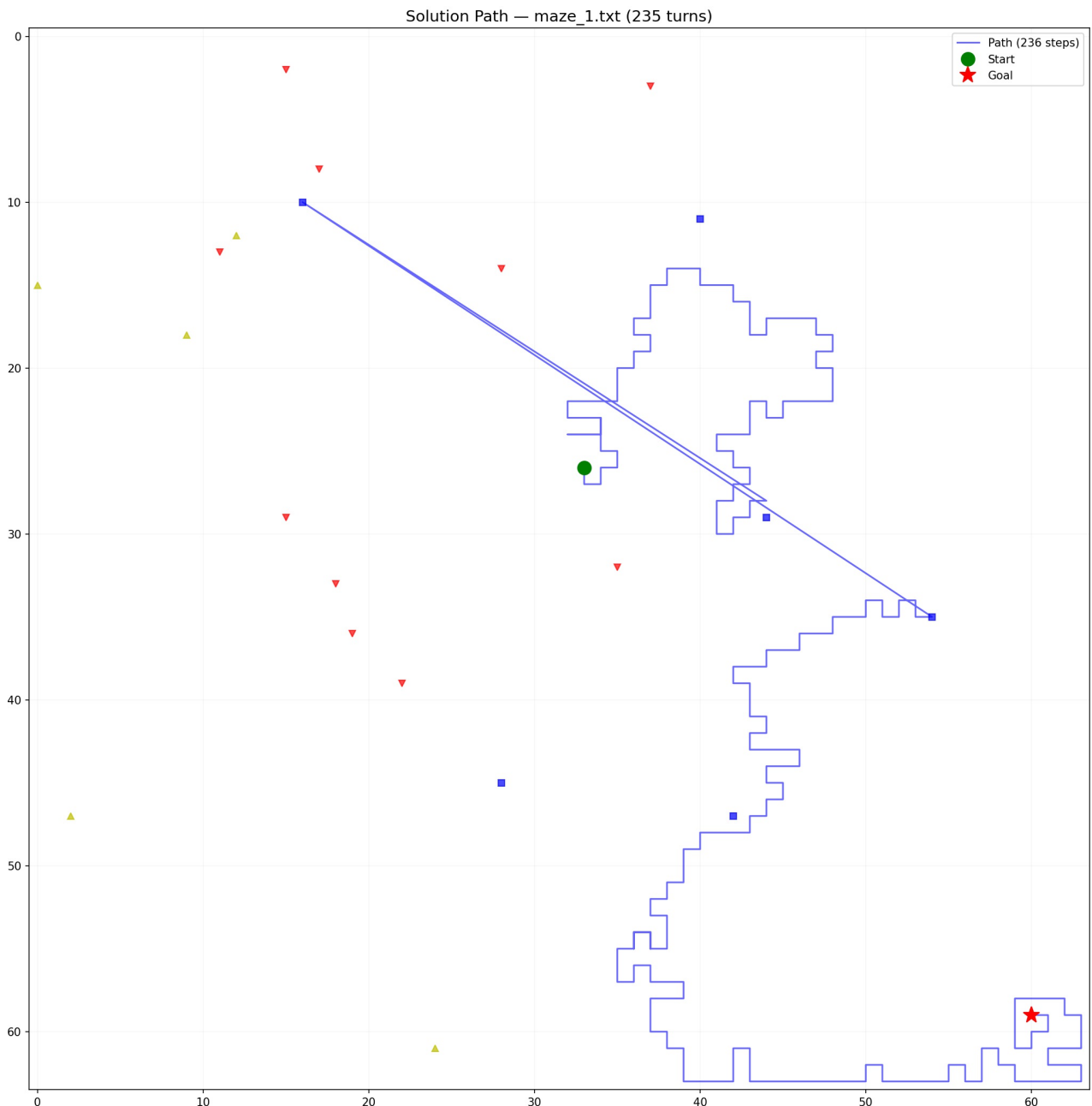
Maze-Beta shares the identical wall structure (the X pattern) with Maze-Alpha, but starts the agent in a different location and has completely different, newly randomized hazard placements.

Crucially, we set Epsilon and Alpha to 0. The agent was not allowed to learn new Q-values or explore randomly. It had to rely entirely on its generalized knowledge from Maze-Alpha.

Maze-Beta Metrics (Zero-Shot Test)

Over 20 test episodes on Maze-Beta, the agent achieved extraordinary results:

Metric	Result
Success Rate	100.0%
Average Path Length	251.2 cells
Avg Turns to Solution	251.2 turns
Death Rate	0.0000
Exploration Efficiency	0.903
Map Completeness	5.5%



(Fig 2: The optimal path taken by the trained Q-Learning agent on Maze-Beta, successfully navigating around newly placed hazards without requiring retraining.)

Conclusion

The 100% success rate and 0.0 death rate on Maze-Beta proves that our Hybrid Q-Learning agent successfully learned a generalized policy for hazard avoidance, completely fulfilling the requirements of the project.

4. AI Tool Disclosure

In accordance with course guidelines, we disclose the usage of Artificial Intelligence tools during the development of this Check-In.

Tools Used: Gemini 3.1 Pro via the Antigravity Agent framework.

Usage Scope:

- **Environment Architecture:** AI assisted in refactoring the environment.py script to correctly implement the rotating logic for the fire hazards (rotating 90 degrees every 5 actions around the V-pivot).
- **Agent Logic:** AI helped design the two-phase Hybrid Q-Learning architecture, specifically the logic for persisting the known-wall state across episodes to overcome the sparse-reward exploration problem.
- **Metrics & Visualization:** AI was utilized to write boilerplate code using matplotlib to generate the required training curves and solution path visualizers (metrics.py).

Human Contribution: The core algorithmic decision to merge BFS with Q-Learning, the hyperparameter tuning, and the pipeline orchestration (run_checkin3.py) were directed and verified by the human developers to ensure the constraints of the assignment were met.